

IB MATH AA SL

Lesson 44

Choosing & Scoping Your IA

IA

Internal Assessment · 60-minute lesson

IB Mathematics IA (20%) · Property of Lynda Badmus Education

Do Now — recall

ON YOUR WHITEBOARD

1. What % of your final grade is the IA?
2. Roughly how long is the IA (pages)?
3. In one word: what makes a great IA topic?

Answers (click to reveal)

1. 20%
2. about 12–20 pages
3. interest (personal engagement)

What we're learning today

LEARNING OBJECTIVES

- 1 Understand**
what the Mathematics IA (the 'exploration') is.
- 2 Choose**
a topic with genuine personal interest.
- 3 Scope**
the task so it is focused, not too big or small.
- 4 Check**
the maths is at the right level (AA SL).

SUCCESS CRITERIA — I CAN...

- ✓
explain the purpose of the IA
- ✓
brainstorm topics linked to my interests
- ✓
narrow a broad idea into a focused question
- ✓
judge whether the maths is suitable for AA SL

What the IA is

THE EXPLORATION

IA = 20% of your final grade

A written exploration

of a maths topic that interests you

Around 12-20 pages

your own investigation, not an essay

Marked

by your teacher against 5 criteria

GOOD vs WEAK

Good

focused, personal, real maths used well

Weak

too broad, copied theory, little of you in it

From interest to topic

QUESTION

A student loves football.

How could that become a focused IA?

SOLUTION

Broad interest: football.

Narrow it: 'Modelling the flight of a free kick.'

Focus it with maths: use a quadratic / projectile model

and compare to real shot data.

Your turn — narrow it

ANSWER IN YOUR BOOK

1. Interest: music. Suggest a focused IA idea.
2. Interest: travel. Suggest a focused IA idea.
3. Why is 'the history of pi' a weak topic?

Answers (click to reveal)

1. e.g. modelling a sound wave with sine functions
2. e.g. optimising a route / cost with functions
3. it's just retelling theory — little personal maths

Scoping it right

NOT TOO BIG, NOT TOO SMALL

focused question + your own data/analysis

Too broad

'calculus' — impossible to do well

Too small

one calculation — not enough depth

Just right

one clear question explored thoroughly

CHECKLIST

Interest

do I actually care about it?

Maths

is it AA SL level?

Doable

can I get data / results?

Judge the scope

QUESTION

Is 'Investigating all types of functions' a good IA scope?

SOLUTION

No — far too broad to explore in depth.

Better: pick ONE context, e.g.

'Which function best models my city's monthly rainfall?'

One focused question, real data, clear analysis.

Exam-style question

A student proposes the IA title: “Statistics.”

- (a) Explain why this title is unsuitable. [2]
- (b) Rewrite it as a focused, personal exploration question. [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) Far too broad / not a question / no personal context (A1)(A1)
- (b) e.g. 'Is there a correlation between hours of sleep and test scores in my class?' (A1)(A1)

Common mistakes

**Too broad**

A whole topic is not an IA — narrow to one question.

**No personal link**

Pick something YOU find interesting.

**Just theory**

Re-explaining a known proof scores poorly — do your own analysis.

**Wrong level**

Keep the maths at AA SL, used correctly.

Mixed practice

QUESTIONS

1. What % is the IA worth?
2. Turn 'sport' into a focused IA idea.
3. Name one sign of a weak topic.
4. Name one thing a good topic needs.

Answers (click to reveal)

1. 20%
2. e.g. modelling a basketball shot's arc
3. too broad / copied theory
4. personal interest + real maths

Plenary & exit ticket

KEY FORMULAS

$$IA = 20\%$$

one focused question

EXIT TICKET — before you leave

1. IA worth what %?
2. Best source of a topic?
3. Broad or focused — which wins?

Answers (click to reveal)

1. 20%
2. your own interests
3. focused

Self-assess:
• need more help
• nearly there
• confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Draft three possible IA questions.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 45

Structuring the IA to the assessment criteria.

Criteria A – E